MILITARY SPECIFICATION

DRILL, BONE, PNEUMATIC

This specification is approved for use for use within the Defense Personnel Support Center and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pneumatic bone drill suitable for shaping and drilling bone material during orthopedic, neurosurgical and maxillofacial surgical procedures. The accessories and burs provided shall be as specified and suitable for the purpose intended.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

GG-I-526	-	Instruments, Dental and Surgical; General Specification.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-601	-	Boxes, Wood, Cleated-plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Cover.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Set-up.

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MIL-P-116	-	Methods of Preservation.
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-B-37959	-	Burs, Surgical, Tungsten Carbide, General Requirements For.
MIL-B-37959/5	-	Bur, Surgical, Tungsten Carbide, 2 mm by 7.5 mm Head, Sidecutting, Tapered, Medium Shank.
MIL-B-37959/7	-	Bur, Surgical, Tungsten Carbide, 1.5 mm, Round Cutting, Medium Shank.
MIL-B-37959/17	-	Bur, Surgical, Tungsten Carbide, 5 mm, Head Diameter, Round Cutting, Medium Shank.
MIL-B-37959/20	-	Bur, Surgical, Tungsten Carbide, 4 mm, Head Diameter, Round Cutting, Medium Shank.
MIL-B-37959/21	-	Bur, Surgical, Tungsten Carbide, 1.5 mm by 4.5 mm Head, Side Cutting, Straight, Medium Shank
MIL-B-37966	-	Burs, Bone Surgery, Wire Pass Drill.

STANDARDS

FEDERAL

FED-STD-151	-	Metals, Test Methods.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads.
MIL-STD-794	-	Parts and Equipment, Procedures for Packaging of.

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(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

D 2240 - Standard Test Method for Rubber Property-Durometer Hardness.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The drill and its accessories and spare parts shall be in accordance with applicable requirements of Federal Specification GG-I-526 unless otherwise specified herein.

3.2 Speed and torque. The drill shall have an operating range of 90,000 to 110,000 revolutions per minute. The nominal torque produced by the drill shall be 1.75 inch-ounces. The minimum available torque shall be 1.50 inch-ounces.

3.3 Control. The drill shall have on/off and variable speed control capabilities.

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3.4 Power source. The drill shall be powered by dry compressed nitrogen or compressed air which has been filtered and is 99.95 percent clean and dry. The drill shall operate off of compressed air having an operating pressure of 90-110 psi.

3.5 Gas consumption. The maximum allowable gas consumption shall not be greater than 8 cubic feet (0.23 cubic meters) per minute in the specified operating range.

3.6 Resistance to autoclaving. The drill shall withstand five autoclavings performed in accordance with 4.4.4 without visible corrosion or impairment of function. Boil test of GG-I-526 shall not apply.

3.7 Accessories. The following accessories shall be provided with the drill.

3.7.1 Nitrogen regulator. Shall be capable of regulating the flow of nitrogen to the drill. The regulator shall have a standard "Schrader" style female connection with a standard Compressed Gas Association (CGA) nitrogen fitting on the input connection. The regulator shall have two gauges. One gauge shall be for operating pressure and a second gauge shall be for supply pressure.

3.7.2 Sterilization rack. Three anodized aluminum racks shall hold up to 20 burs each. The racks shall include handle for ease of transportation and shall be capable of sustaining repeated steam sterilizations without visible corrosion or impairment of function. The racks shall be tested for resistance to autoclaving in accordance with 4.4.4.

3.7.3 Bur cleaning brush. The brush shall have a two inch long plastic handle and be 3/8 inch in diameter. The brush shall have 0.005 inch diameter, one-half inch length, stainless steel wires protruding from one end of the handle.

3.7.4 Medium size bur guard.

3.7.4.1 Style, design and dimensions. Shall be a medium size bur guard suitable for use in stabilizing medium burs during orthopedic, oral, plastic and reconstructive surgery. The guard shall attach directly to the drill. The guard shall conform to Figure 1 for style, design and dimensions.

3.7.4.2 Material. The guard shall be made from Type 17-4 precipitation hardening corrosion-resisting steel. The Rockwell hardness shall be between C-40 and C-44.

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3.7.4.3 Resistance to autoclaving. The guard shall be able to sustain repeated autoclaving without visible corrosion or impairment of function. The guard shall be tested in accordance with 4.4.4.

3.7.5 Long bur guard.

3.7.5.1 Style, design and dimensions. Shall be a long size bur guard suitable for use in stabilizing long burs during orthopedic, oral, plastic and reconstructive surgery. The guard shall attach directly to the drill. The guard shall conform to Figure 2 for style, design and dimensions.

3.7.5.2 Material. The guard shall be made from Type 17-4 precipitation hardening corrosion-resisting steel. The Rockwell hardness shall be C-40 to C-44.

3.7.5.3 Resistance to autoclaving. The guard shall be able to sustain repeated autoclaving without visible corrosion or impairment of function. The guard shall be tested in accordance with 4.4.4.

3.7.6 Bur guard lubricant. The lubricant shall be one ounce of SAE 20-weight motor oil. The lubricant shall be supplied in a plastic container with an applicator tip and cap.

3.7.7 Instrument case.

3.7.7.1 Style, design and dimensions. Shall be a perforated case large enough to hold the drill, hose and other accessories. The case shall be made from a suitable corrosion-resisting material that is capable of sustaining repeated steam sterilization.

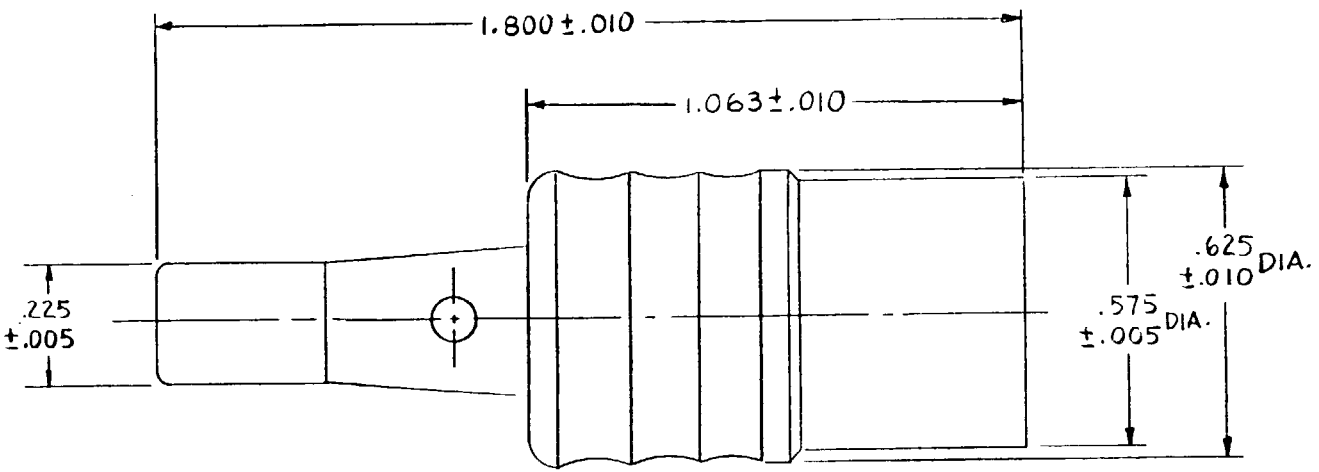
3.7.7.2 Resistance to autoclaving. The instrument case shall be able to sustain repeated autoclaving without visible corrosion or impairment of function. The instrument case shall be tested in accordance with 4.4.4.

3.7.8 Hose. Shall be a dual purpose hose with the supply hose internal to the exhaust hose. The supply end of the hose shall have a male standard "Schrader" adapter end. The discharge end shall have a quick disconnect connection that mates with the connection on the drill. The hose shall be approximately 10 feet long. Two hoses shall be supplied with each drill.

3.7.8.1 Supply hose. The supply hose shall be of dual wall construction. The inner tube shall be made from neoprene rubber and the outer wrap shall be black nylon braid. The hose shall have a minimum burst pressure of 500 psi.

3.7.8.2 Discharge hose. Shall be a high strength silicone rubber hose with a minimum Durometer hardness of 60. The minimum burst pressure shall be 50 psi.

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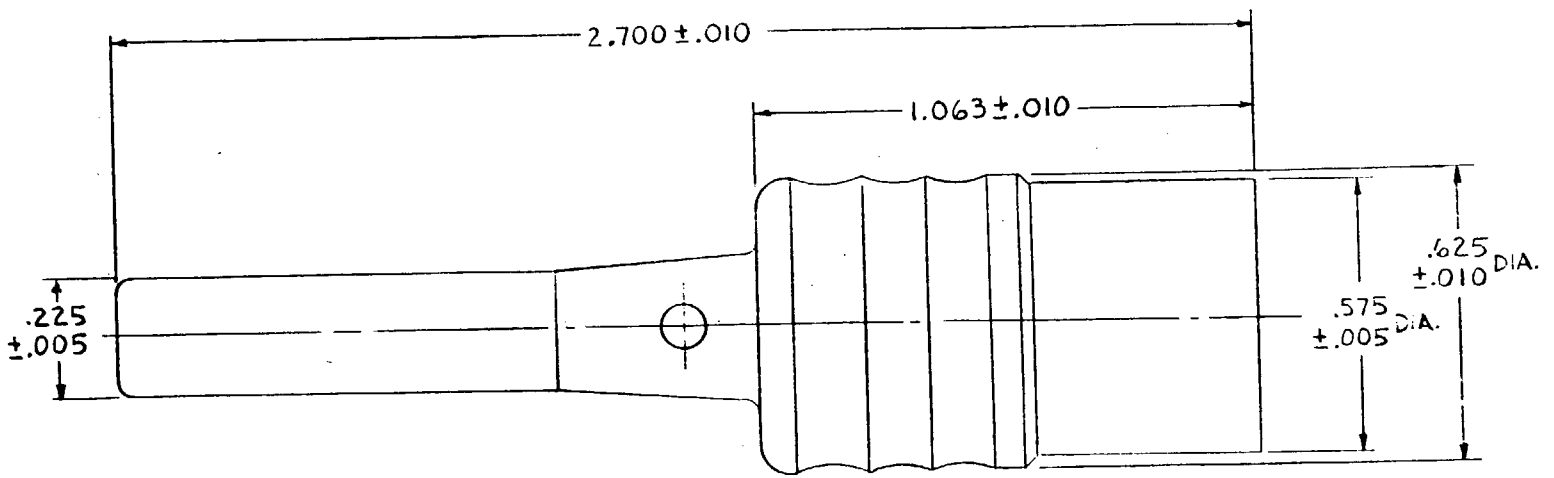


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FIGURE 1

BURGUARD ASSEMBLY, MEDIUM

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FIGURE 2

BURGUARD ASSEMBLY, LONG

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3.7.8.3 Resistance to autoclaving. The supply hose and the discharge hose shall be able to sustain repeated autoclaving without corrosion or impairment of function. The supply hose and the discharge hose shall be tested in accordance with 4.4.4.

3.8 Spare parts. The drill shall be provided with the following spare parts.

3.8.1 Side cutting, straight. Shall be a side cutting, straight shape, medium shank, surgical bur and shall be in accordance with Military Specification MIL-B-37959/21(DM).

3.8.2 Side cutting, tapered. Shall be a side cutting, tapered shape, medium shank, surgical bur and shall be in accordance with Military Specification MIL-B-37959/5(DM).

3.8.3 Round cutting, 1.5 mm. Shall be a round cutting, medium shank, surgical bur and shall be in accordance with Military Specification MIL-B-37959/7(DM).

3.8.4 Round cutting, 3.0 mm. Shall be a round cutting, medium shank, surgical bur with a head and shank. The head shall be made of C1 or C2 Grade tungsten carbide with a grain size of 1 to 2 microns. The shank shall be made of C13 to C14 tungsten carbide. The bur shall be examined at 7X magnification for bur defects not readily perceptable to the naked eye. The proximal end shall be round and non-notched. The bur shall be of the design and dimensions shown in Figure 3.

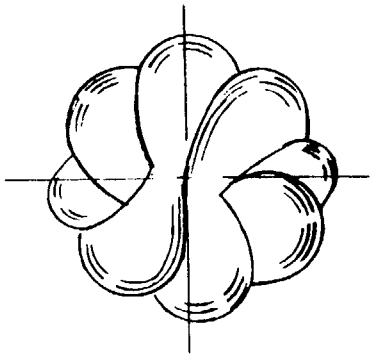
3.8.5 Round cutting, 4.0 mm. Shall be a round cutting, medium shank, surgical bur and shall be in accordance with Military Specification MIL-B-37959/20(DM).

3.8.6 Round cutting, 5.0 mm. Shall be a round cutting, medium shank, surgical bur and shall be in accordance with Military Specification MIL-B-37959/17(DM).

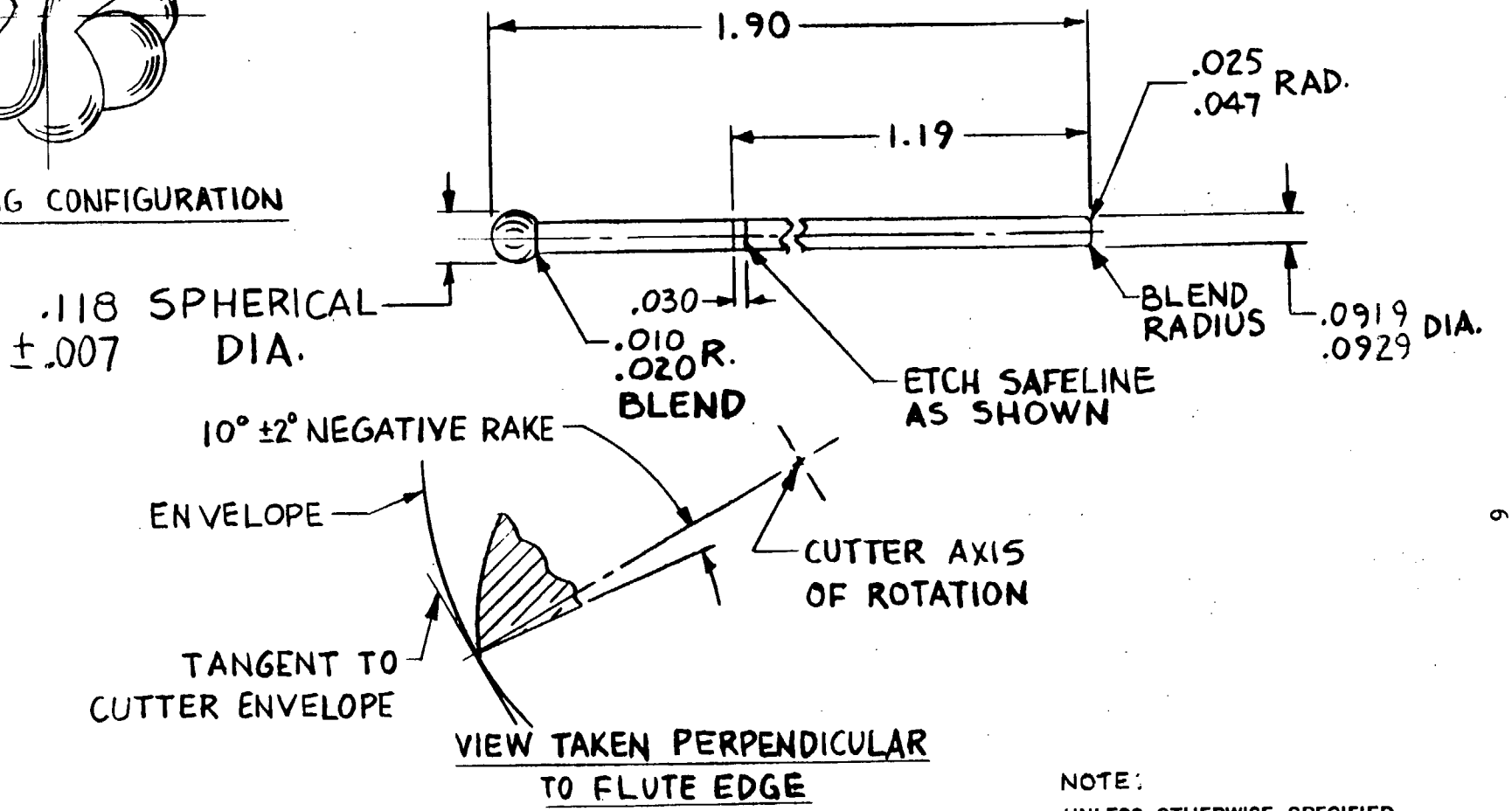
3.8.7 Round cutting, 6.5 mm. Shall be a round cutting, medium shank, surgical bur with a head and shank. The head shall be made of C1 or C2 Grade tungsten carbide with a grain size of 1 to 2 microns. The shank shall be made of C13 to C14 tungsten carbide. The bur shall be examined at 7X magnification for bur defects not readily perceptable to the naked eye. The proximal end shall be round and non-notched. The bur shall be of the design and dimensions shown in Figure 4.

3.8.8 Wire pass drill, 1.0 mm. Shall be a wire pass drill, medium shank, surgical bur, 1.0 mm in diameter and shall be in accordance with Military Specification MIL-B-37966(DM).

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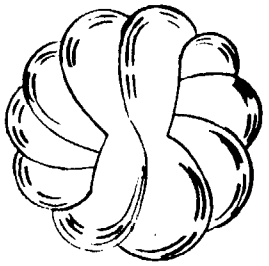
END CUTTING CONFIGURATION



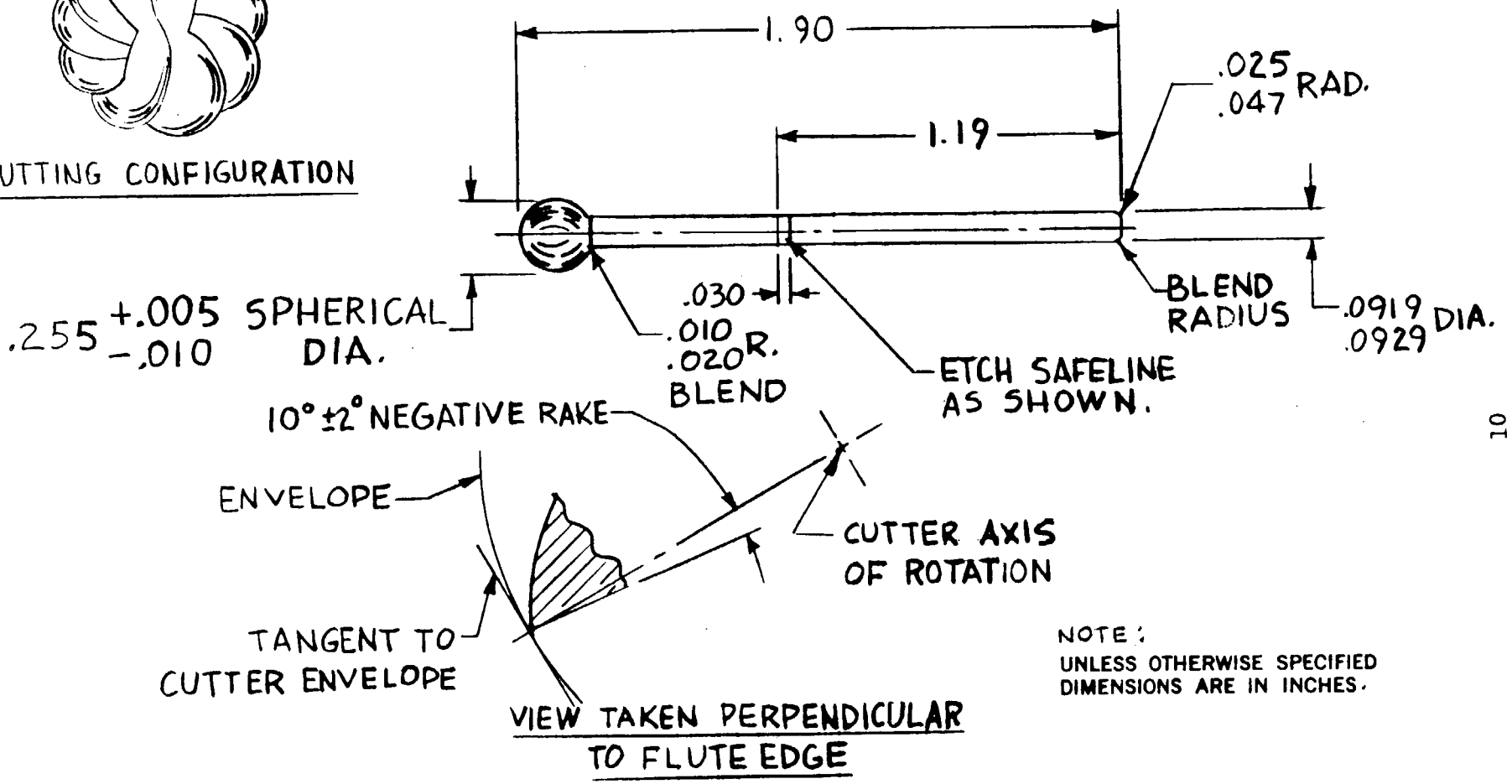
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FIGURE 3
BUR, 3MM ROUND, CUTTING

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END CUTTING CONFIGURATION



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FIGURE 4
BUR, 6.5MM ROUND, CUTTING

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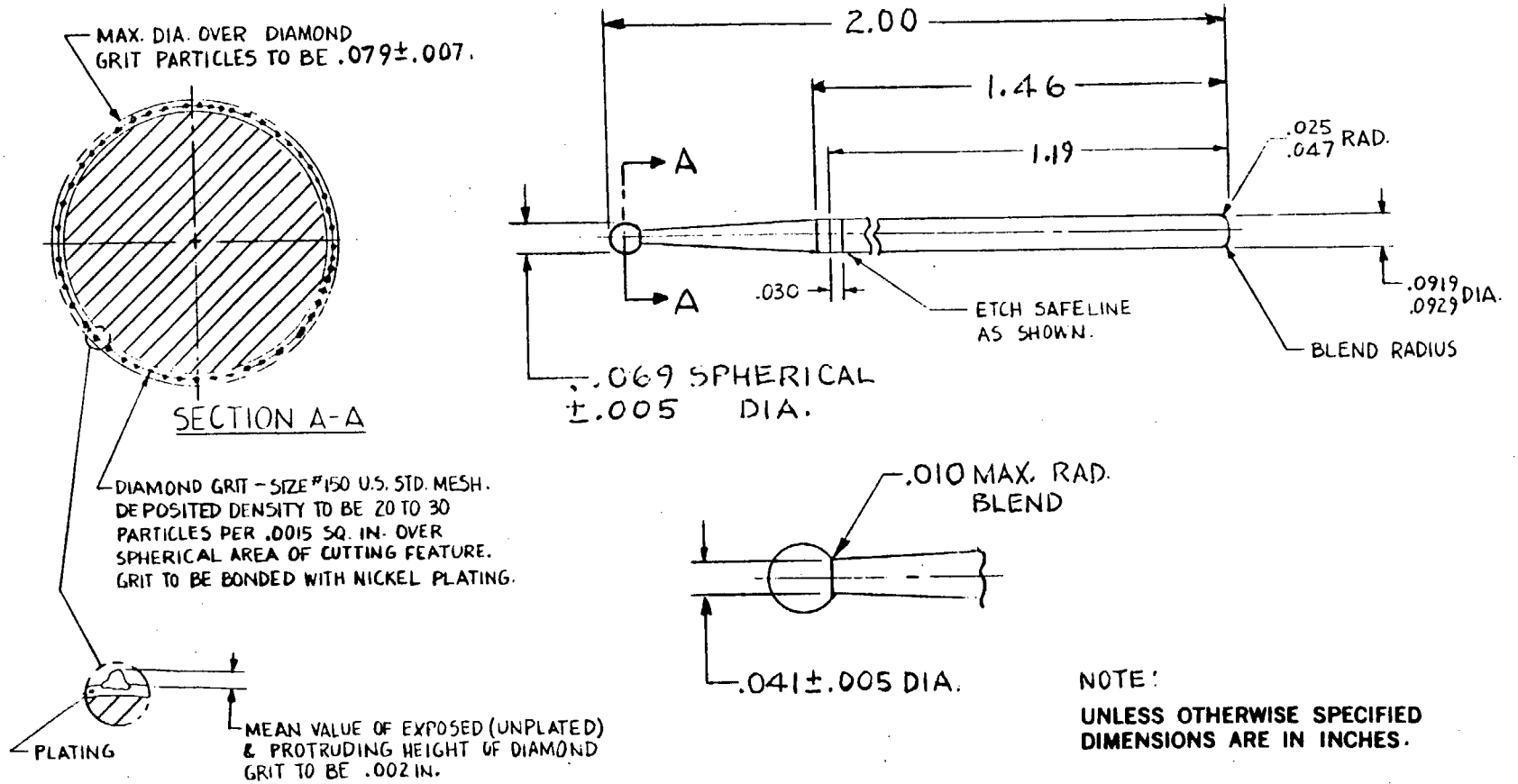


FIGURE 5
BUR, ROUND DIAMOND, 2.0 MM

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3.8.9 Wire pass drill, 1.5 mm. Shall be a wire pass drill, medium shank, surgical bur, 1.5 mm in diameter and shall be in accordance with Military Specification MIL-B-37966(DM).

3.8.10 Wire pass drill, 2.0 mm. Shall be a wire pass drill, medium shank, surgical bur, 2.0 mm in diameter and shall be in accordance with Military Specification MIL-B-37966(DM).

3.8.11 Round diamond, 2.0 mm. Shall be a spherically shaped, diamond coated bur with a medium shank. The bur shall be in accordance with Figure 5 for style, design and dimensions. The shank and base material for the bur shall be made entirely from Type 17-4 precipitation hardened steel. The Rockwell hardness of the bur shall be C40 to C44. The bur shall have a size #150, US Standard mesh diamond grit bonded to the spherical cutting end of the bur. The deposited density of the diamond grit shall be 20 to 30 particles per square millimeter. The grit is to be bonded to the bur with nickel plating. The mean value of the protruding unplated diamond grit shall be 0.002 inches.

3.8.12 Round diamond, 4.0 mm. Shall be a spherically shaped diamond coated bur with a medium shank. The bur shall be in accordance with Figure 6 for style, design and dimensions. The shank and base material for the bur shall be made entirely from Type 17-4 precipitation hardened steel. The Rockwell hardness of the bur shall be C40 to C44. The bur shall have a size #150, US Standard mesh diamond grit bonded to the spherical cutting end of the bur. The deposited density of the diamond grit shall be 20 to 30 particles per square millimeter. The grit is to be bonded to the bur with nickel plating. The mean value of the protruding unplated diamond grit shall be 0.002 inches.

3.8.13 Round diamond, 6.0 mm. Shall be a spherically shaped diamond coated bur with a medium shank. The bur shall be in accordance with Figure 7 for style, design and dimensions. The shank and base material for the bur shall be made entirely from Type 17-4 precipitation hardened steel. The Rockwell hardness of the bur shall be C40 to C44. The bur shall have a size #150, US Standard mesh diamond grit bonded to the spherical cutting end of the bur. The deposited density of the diamond grit shall be bonded to the bur with nickel plating. The mean value of the protruding unplated diamond grit shall be 0.002 inches.

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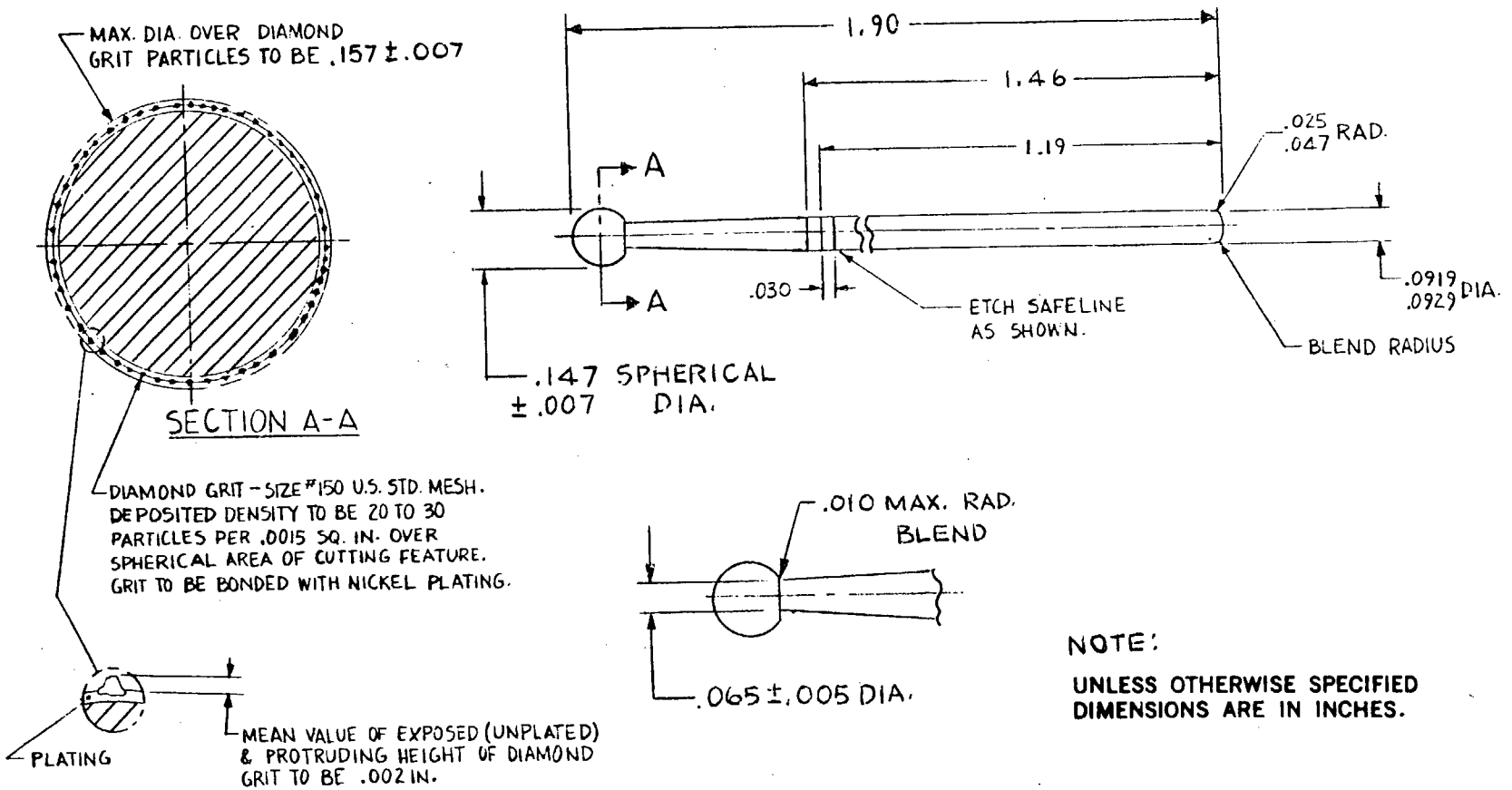
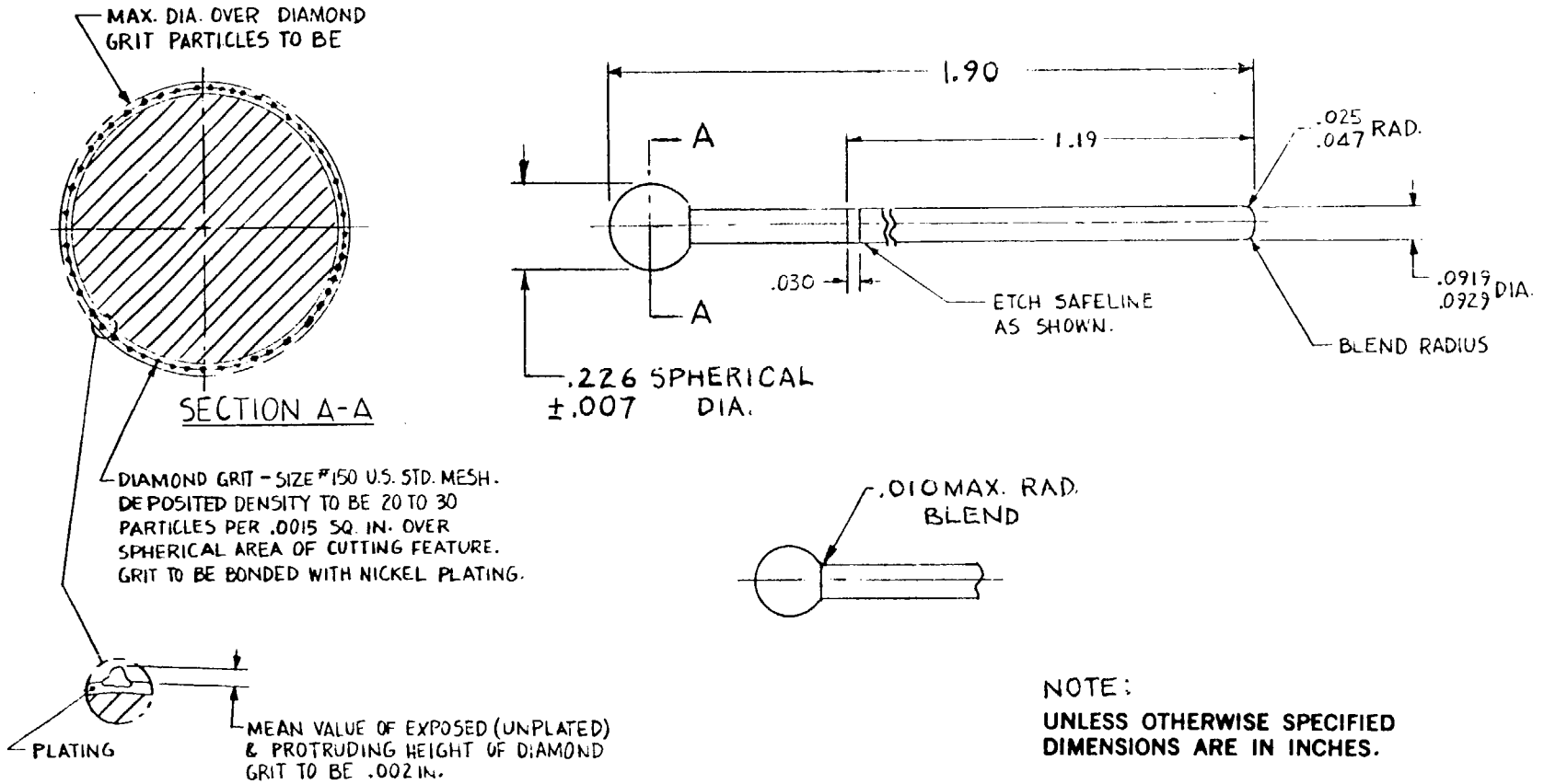


FIGURE 6
BUR, ROUND DIAMOND, 4.0 MM

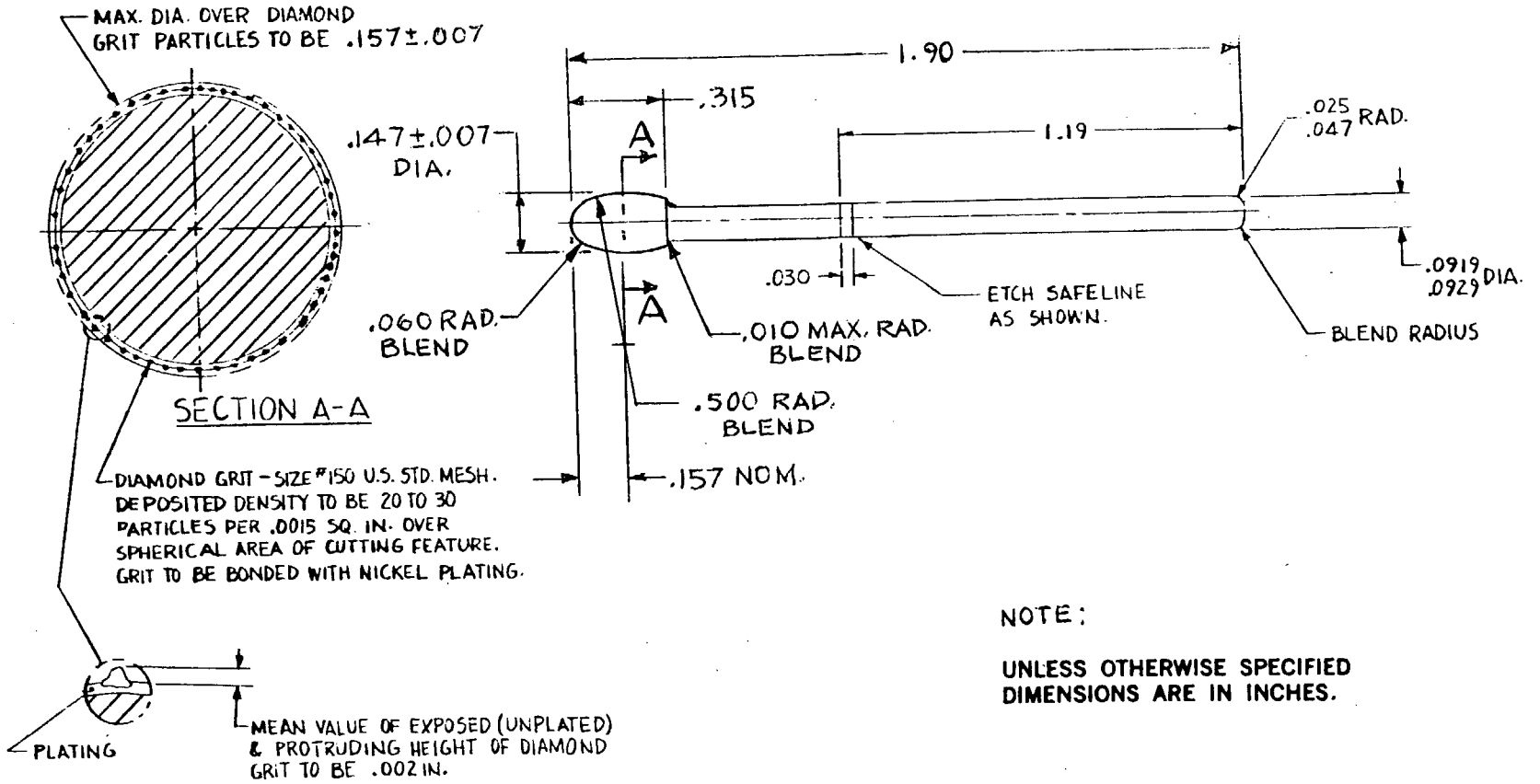
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FIGURE 7
BUR, ROUND DIAMOND, 6.0 MM

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FIGURE 8
BUR, OVAL DIAMOND, 4.0MM x 8.0MM

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3.8.14 Oval diamond, 4.0 mm. Shall be an oval shaped diamond coated bur with a medium shank. The bur shall be in accordance with Figure 8 for style, design and dimensions. The shank and base material for the bur shall be made entirely from Type 17-4 precipitation hardened steel. The Rockwell hardness of the bur shall be C40 to C44. The bur shall have a size #150, US Standard mesh diamond grit bonded to the spherical cutting end of the bur. The deposited density of the diamond grit shall be bonded to the bur with nickel plating. The mean value of the protruding diamond grit shall be 0.002 inches.

3.9 Resistance to autoclaving. The burs shall be able to sustain repeated autoclaving without visible corrosion or impairment of function. The burs shall be tested in accordance with 4.4.4.

3.10 Runout. The total indicated runout of the burs shall not exceed .001 inch when measured as specified in 4.3.2.

3.11 Finish. The finish of the bur shall be satin. The operative end shall shall have a fine honed finish with sharp cutting edges. The safeline shall have an etched finish.

3.12 Workmanship. The burs shall be clean and free from any defect which detract their performance or appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties), of the item.

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4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificate of quality. Certificate of quality supplied by the manufacturer of the steel may be furnished in lieu of actual performance of composition testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Inspection lot. An inspection lot, for purpose of sampling and testing in accordance with MIL-STD-105 shall consist of the bone drill and its accessories and spare parts.

4.2 Quality conformance inspection. The examination required to assure conformance of the bone drill and its accessories and spare parts to be offered for acceptance are classified as specified in Table I and III (see 4.2.1 and 4.3). The tests required for acceptance are specified in Table III of 4.2.2.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The acceptance quality levels (AQLS) shall be as shown in Table I. The unit of product for examination purpose shall be one bone drill and its accessories and spare parts.

TABLE I. Sampling criteria for examination

Type of defect	Inspection level	AQL (percent defective)
Major	I	1.0
Minor	I	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one bone drill and its accessories and spare parts. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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TABLE II. Sampling criteria for tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Drill	speed	3.2	4.4.1	S-1	*
	torque	3.2	4.4.2	S-1	*
	gas consumption	3.5	4.4.3	S-1	*
	autoclaving	3.6	4.4.4	S-1	*
Burs	autoclaving	3.9	4.4.4	S-1	*
	chemical composition	3.8	4.4.6	S-1	*
	hardness	3.8	4.4.5	S-1	*
	runout	3.10	4.3.2	S-1	*
Hoses	chemical composition	3.8	4.4.6	S-1	*
	hardness	3.8	4.4.5	S-1	*
Guard	autoclaving	3.7.4.3 3.7.5.3	4.4.4	S-1	*
	chemical composition	3.7.4.2 3.7.5.2	4.4.6	S-1	*
	hardness	3.7.4.2 3.7.5.2	4.4.5	S-1	*
Sterilization rack	Autoclaving	3.7.2	4.4.4	S-1	*

*AQL for all samples tested is zero.

4.2.3 Raw material. A sufficient quantity of each lot of raw material used in the manufacture bone drill and is accessories and spare parts shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample shall be used for the determination.

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4.3 Examination. The bone drill and its accessories and spare parts shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with the following classification of defects listed in Tables I and III. Examination shall not be restricted to the classified defects shown in Table III. Any item in the sample containing one or more defects shall be rejected and if the number of defective forceps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Style and design not as specified.
102	Bur size or head shape not as specified.
103	Total indicated runout exceeds 0.001 inch.
104	Head diameter exceeds specification requirements at maximum cross section.
105	Etched safeline missing.
106	Operative end and shank dimensions not as specified (except for 104)
107	Hose connections not as specified.
108	Nitrogen gauge not as specified.
Minor	
201	Cutoff end on shank not as shown in applicable figure.
202	Burs not free from defects such as chipped blades, cracks, or pits.
203	Finish on bur not as specified.
204	Overall length of bur not within specification limits.
205	Instructions for use inaccurate, inadequate or missing.
206	Bur rack not as specified.
207	Bur cleaning brush not as specified.
208	Bur guard lubricant not as specified.

4.3.1 Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instruments of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with the gages.

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4.3.2 Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360 degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.4 Tests.

4.4.1 Speed. The drill shall be tested to determine the number of revolutions per minute of the collet at maximum speed. The collet shall rotate at 90,000 to 110,000 rpm. The drill shall be tested both before and after the autoclaving test specified in 4.4.4. Failure to reach the minimum speed requirement either before or after autoclaving shall constitute failure to the speed requirement.

4.4.2 Torque. The drill shall be tested to determine the torque output of the drill. The stall torque, torque required to stall the drill, shall be a minimum of 1.5 inch-ounces. The drill shall be tested before and after the autoclaving test specified in 4.4.4. Failure to deliver the required torque before or after autoclaving shall constitute failure of their requirement.

4.4.3 Gas consumption. The amount of gas consumed, either nitrogen or compressed air, as specified in 3.5, shall be measured while performing the speed and torgue test specified in 4.4.1 and 4.4.2. The gas used to power the drill shall be between 90 to 110 pounds per square inch. The maximum allowable gas consumption shall be eight cubic feet per minute.

4.4.4 Autoclaving. The test samples shall be cleaned in accordance with manufacturer's instructions. The samples shall then b autoclaved for five cycles at 270 $\pm$ 5° for a minimum of 20 minutes. The sample shall be dry heated for a minimum of eight minutes after each autoclaving.

4.4.5 Hardness. The hardness of the burs, bur guards and the drills shall be determined in accordance with ASTM E18 or E92. Hardness of the hose shall be determined in accordance with D2240.

4.4.6 Chemical composition. In the absence of certificates of quality for the raw material, the samples, selected as specified in 4.2.3, shall be analyzed in accordance with Method 111 or 112 of Federal Standard No. 151. In case of dispute, Method 111 shall be used for referee analysis.

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4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The drill along with accessories and spare parts shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. The drill along with the accessories and instructions shall be positioned within the instrument case. The case shall be packaged in accordance with MIL-P-116, Method IC-3. The spare parts shall be wrapped in neutral tissues or other suitable material used and packaged in accordance with MIL-P-116, Method IC-1 or IC-3. The packaged case and repair parts shall be further packaged in a snug fitting box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. The drill along with the accessories and instructions positioned within the carrying case along with the spare parts shall conform to the requirements of MIL-STD-749 for Level C.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Six unit packages shall be packed in an exterior container designed for a type 2 load and conforming PPP-B-601, overseas type, PPP-B-621, class 2, or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

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5.2.2 Level B. Six unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1, or PP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. Six unit packages shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A, B. Pneumatic bone drills along with spare parts specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Pneumatic bone drills along with spare parts packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3. Marking.

5.3.1 Levels A, B, and C. Each unit pack, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

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6. NOTES

6.1 Intended use. The pneumatic bone drill shall be used in drilling and shaping bone material during orthopedic, neurosurgical and maxillofacial surgical procedures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.3 National Stock Number (NSN) coverage. This specification covers the following National Stock Number:

National Stock No.	Item Identification
6515-01-204-2664	DRILL, BONE, PNEUMATIC, Nitrogen or Compressed Air

6.4 National Stock Number (NSN) inclusion. This specification references the following National Stock Numbers:

National Stock No.	Item Identification
6515-01-139-7729	BUR, SURGICAL, TUNGSTEN CARBIDE, 1.5 mm By 4.5 mm Head, Side Cutting, Straight, Medium Shank
6515-01-139-7730	BUR, SURGICAL, TUNGSTEN CARBIDE, 2 mm By 7.5 mm Head, Side Cutting, Tapered, Medium Shank
6515-01-139-7732	BUR, SURGICAL, TUNGSTEN CARBIDE, 1.5 mm, Round Cutting, Medium Shank
6515-01-139-7734	BUR, SURGICAL, TUNGSTEN CARBIDE, 5 mm Head Diameter, Rounding Cutting, Medium Shank
6515-01-139-7736	BUR, SURGICAL, TUNGSTEN CARBIDE, 4 mm, Round Cutting, Medium Shank
6515-01-139-7741	BUR, BONE SURGERY, Wire Pass Drill, 1.5 mm Head Diameter

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National Stock No.	Item Identification
6515-01-139-7744	BUR, BONE SURGERY, Wire Pass Drill, 1.0 mm Head Diameter
6515-01-139-7745	BUR, BONE SURGERY, Wire Pass Drill, 2.0 mm Head Diameter

6.5 Subject term (key word) listing.

Bone drill
Drill
Pneumatic bone drill

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3939

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-42034(DM)	Drill, Bone, Pneumatic

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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